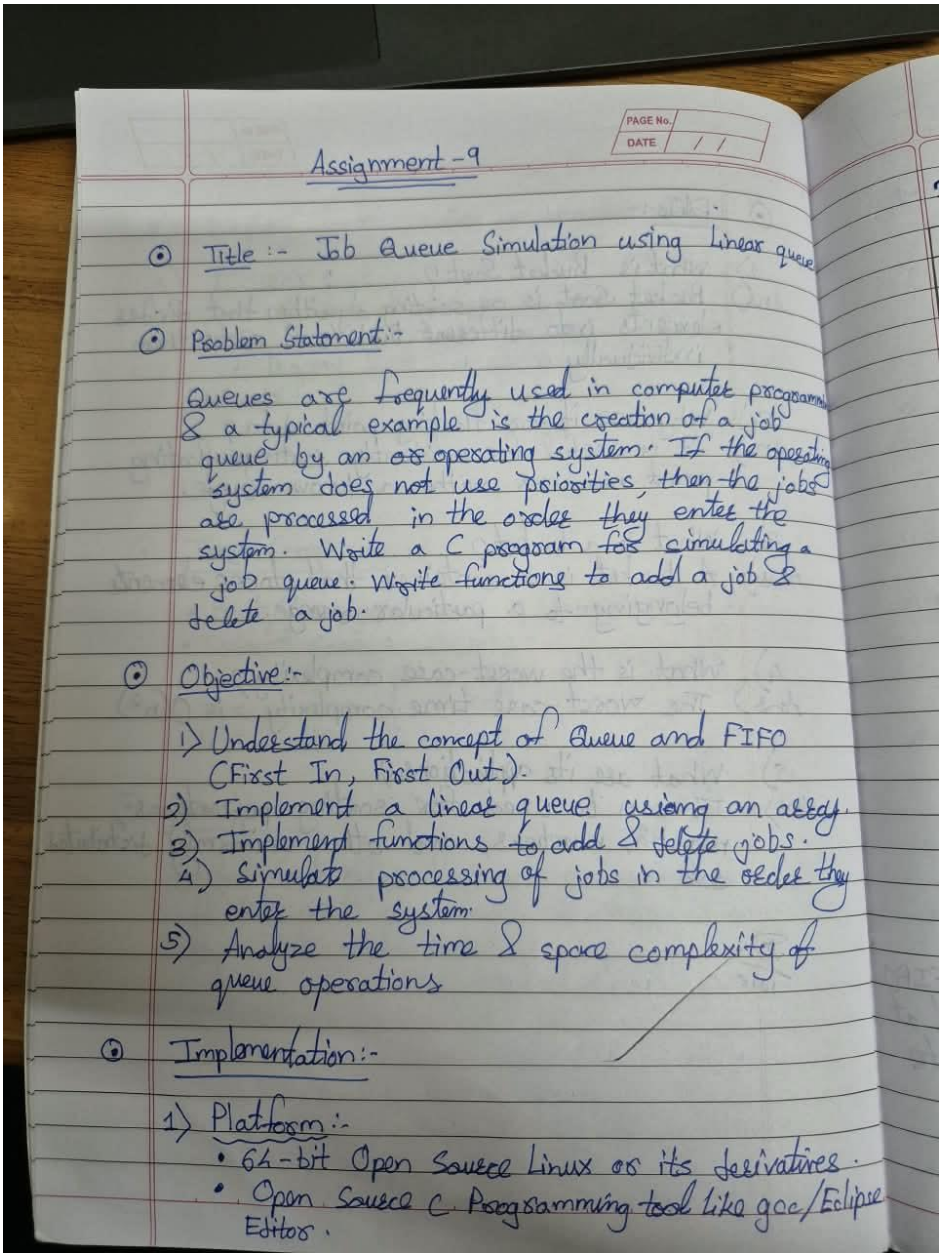
ASSIGNMENT-9

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BATCH - A1



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⊙ Title :- Job Queue Simulation using Linear queue

⊙ Problem Statement :-

Queues are frequently used in computer programming & a typical example is the creation of a job queue by an ~~os~~ operating system. If the operating system does not use priorities, then the jobs are processed in the orders they enter the system. Write a C program for simulating a job queue. Write functions to add a job & delete a job.

⊙ Objective :-

1) Understand the concept of Queue and FIFO (First In, First Out).
2) Implement a linear queue using an array.
3) Implement functions to add & delete jobs.
4) Simulate processing of jobs in the order they enter the system.
5) Analyze the time & space complexity of queue operations

⊙ Implementation :-

1) Platform :-
- 64-bit Open Source Linux or its derivatives.
- Open Source C Programming tool like gcc/Eclipse Editor.

2) Test Conditions:-

Test Case	Input	Expected Output
TC01	Add one job	Job added successfully.
TC02	Add multiple jobs.	Jobs stored in FIFO order.
TC03	Delete a job	First entered job is deleted / processed.
TC04	Delete from empty queue.	Queue underflow message.
TC05	Add job when queue is full.	Queue overflow message.
TC06	Display pending jobs.	All jobs displayed from front to rear.

3) Pseudocode:-

START

Initialize front = -1, rear = -1

ADD JOB:
If rear = MAX - 1 → Queue Overflow
Else read job
If front = -1 → front = 0
rear = rear + 1
queue[rear] = job

DELETE JOB:
If front = -1 OR front > rear → Queue Underflow
Else display & delete queue[front]
front = front + 1

If front > rear → front = rear = -1

DISPLAY JOBS:
If front = -1 → Queue Empty
Else display jobs from front to rear.

Repeat menu until user selects Exit.

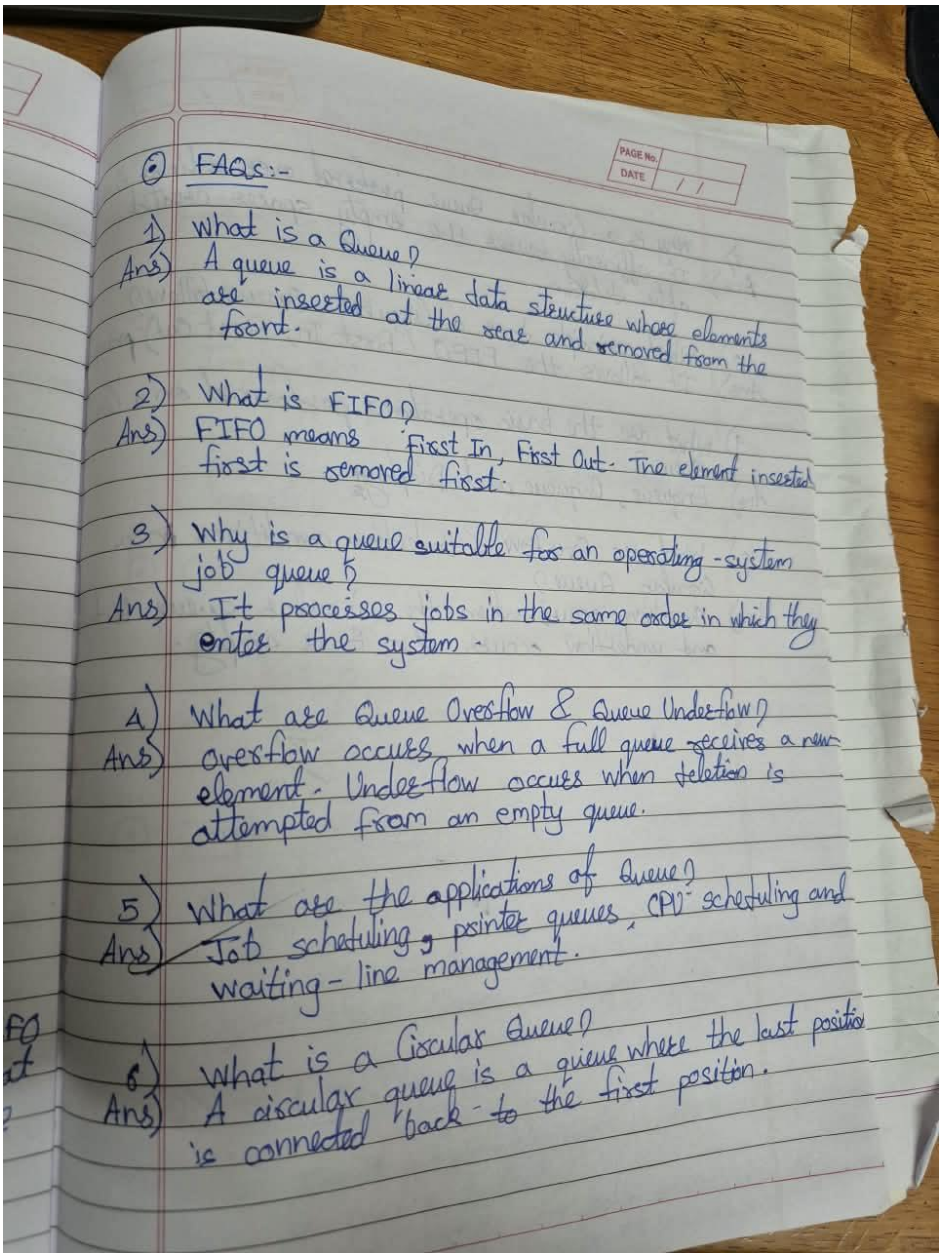
END.

4) Time Complexity :-

Operation	Time Complexity
(i) Add Job (Enqueue)	O(1)
(ii) Delete Job (Dequeue)	O(1)
(iii) Display Jobs	O(n)

⊙ Conclusion:-

The job queue simulation demonstrates the FIFO principle of a linear queue. Jobs are added at the rear and processed / deleted from the front in the same order in which they enter the system.



⊙ FAQs:-

1) What is a Queue?
Ans) A queue is a linear data structure where elements are inserted at the rear and removed from the front.

2) What is FIFO?
Ans) FIFO means First In, First Out. The element inserted first is removed first.

3) Why is a queue suitable for an operating-system job queue?
Ans) It processes jobs in the same order in which they enter the system.

4) What are Queue Overflow & Queue Underflow?
Ans) Overflow occurs when a full queue receives a new element. Underflow occurs when deletion is attempted from an empty queue.

5) What are the applications of Queue?
Ans) Job scheduling, printer queues, CPU scheduling and waiting-line management.

6) What is a Circular Queue?
Ans) A circular queue is a queue where the last position is connected back to the first position.

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7) Why is a Circular Queue preferred over a Linear Queue?
Ans) It efficiently reuses the empty spaces created after deletion.

8) What principle does a Circular Queue follow?
Ans) It follows the FIFO (First In, First Out) principle.

9) What are the basic operations performed on a Circular Queue?
Ans) Enqueue, Dequeue and Display.

10) What are Overflow & Underflow conditions in a Circular Queue?
Ans) Overflow occurs when the circular queue is full, and underflow occurs when it is empty.

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